

Climate Change Governance: Science, Data & Models



Lecture 4: Radiative forcing and feedbacks

ENGG562/ENV562 – Fall 2026

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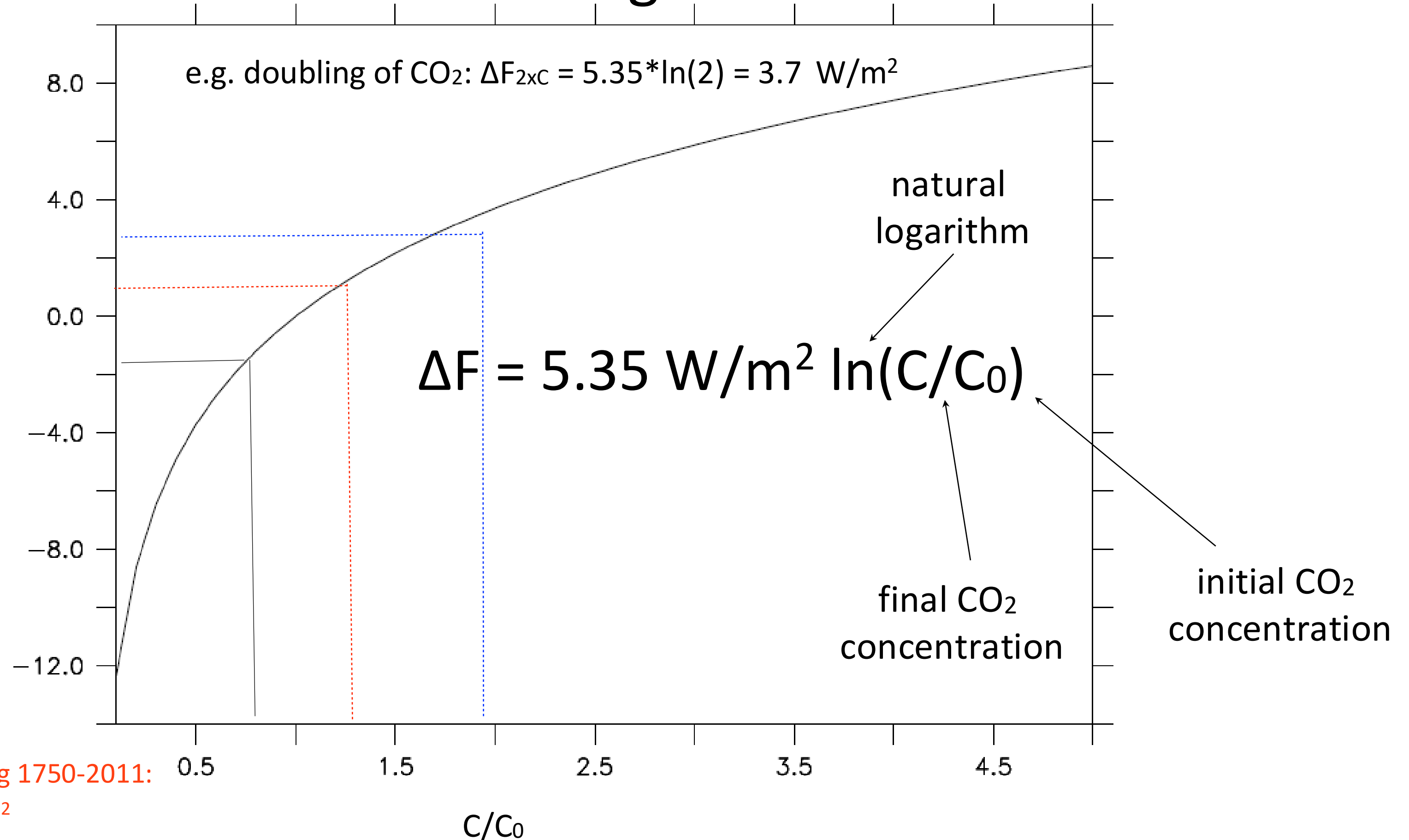
Radiative forcing as driver of Climate Change

“The change in the net, downward minus upward, radiative flux (expressed in W/m^2) due to a change in an external driver of climate change, such as a change in the concentration of carbon dioxide (CO_2), the concentration of volcanic aerosols or the output of the Sun.” – IPCC AR6

Sign convention:

- More energy coming back to earth → positive forcing leads to warming
- Less energy coming back to earth → negative forcing leads to cooling.

Radiative Forcing for CO₂:



Model Input

CO₂ (ppm)

400

CH₄ (ppm)

1.7

Trop. Ozone (ppb)

28

Strat. Ozone scale

1

Water Vapor Scale

1

Freon Scale

1

Temperature Offset, C

0

Locality

Tropical Atmosphere

No Clouds or Rain

Altitude (km)

70

Looking down

Save This Run to Background

Show Raw Model Output

Model Output

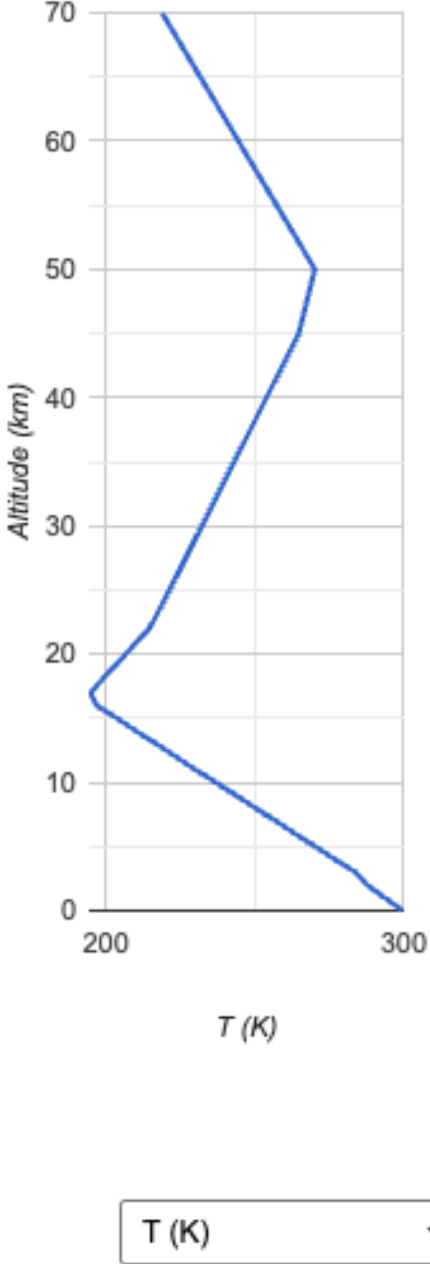
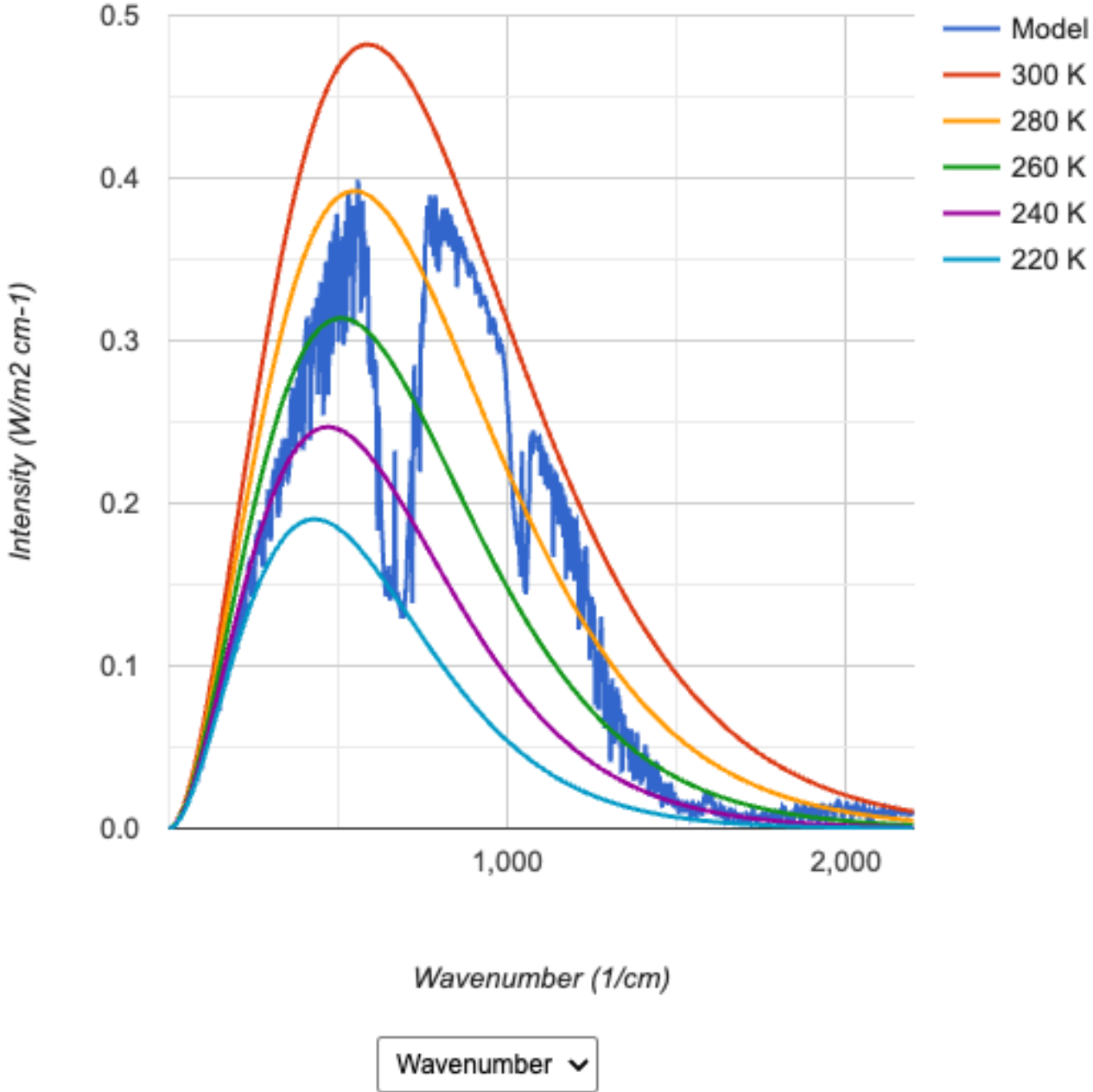
Upward IR Heat Flux

298.52 W/m²

Ground Temperature

299.7 K

Spectrum expanded 5-11-17, changing the IR out value.



Model Input

CO₂ (ppm)

CH₄ (ppm)

Trop. Ozone (ppb)

Strat. Ozone scale

Water Vapor Scale

Freon Scale

Temperature Offset, C

Locality

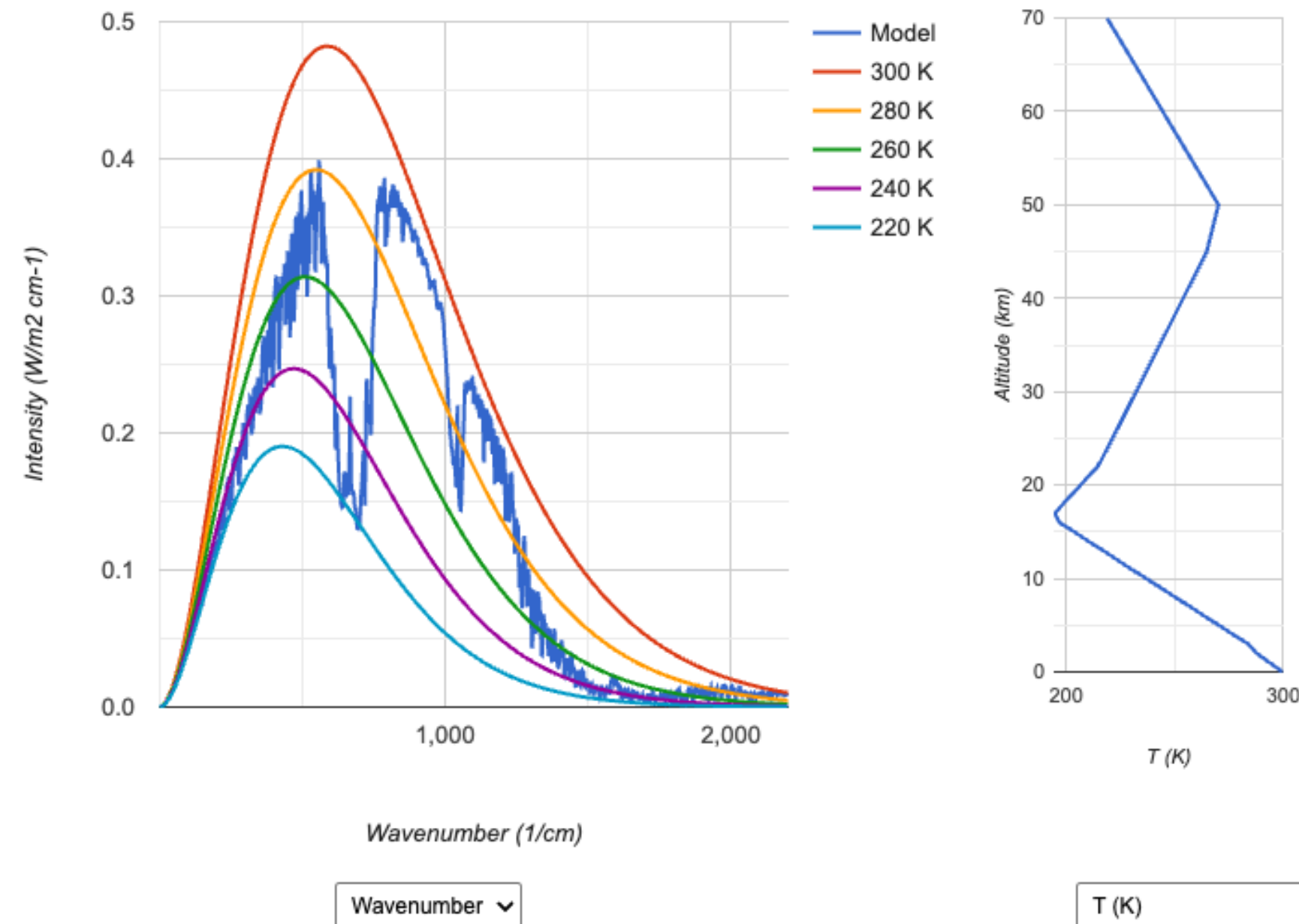
Altitude (km)

Model Output

Upward IR Heat Flux **295.191** W/m²

Ground Temperature **299.7** K

Spectrum expanded 5-11-17, changing the IR out value.



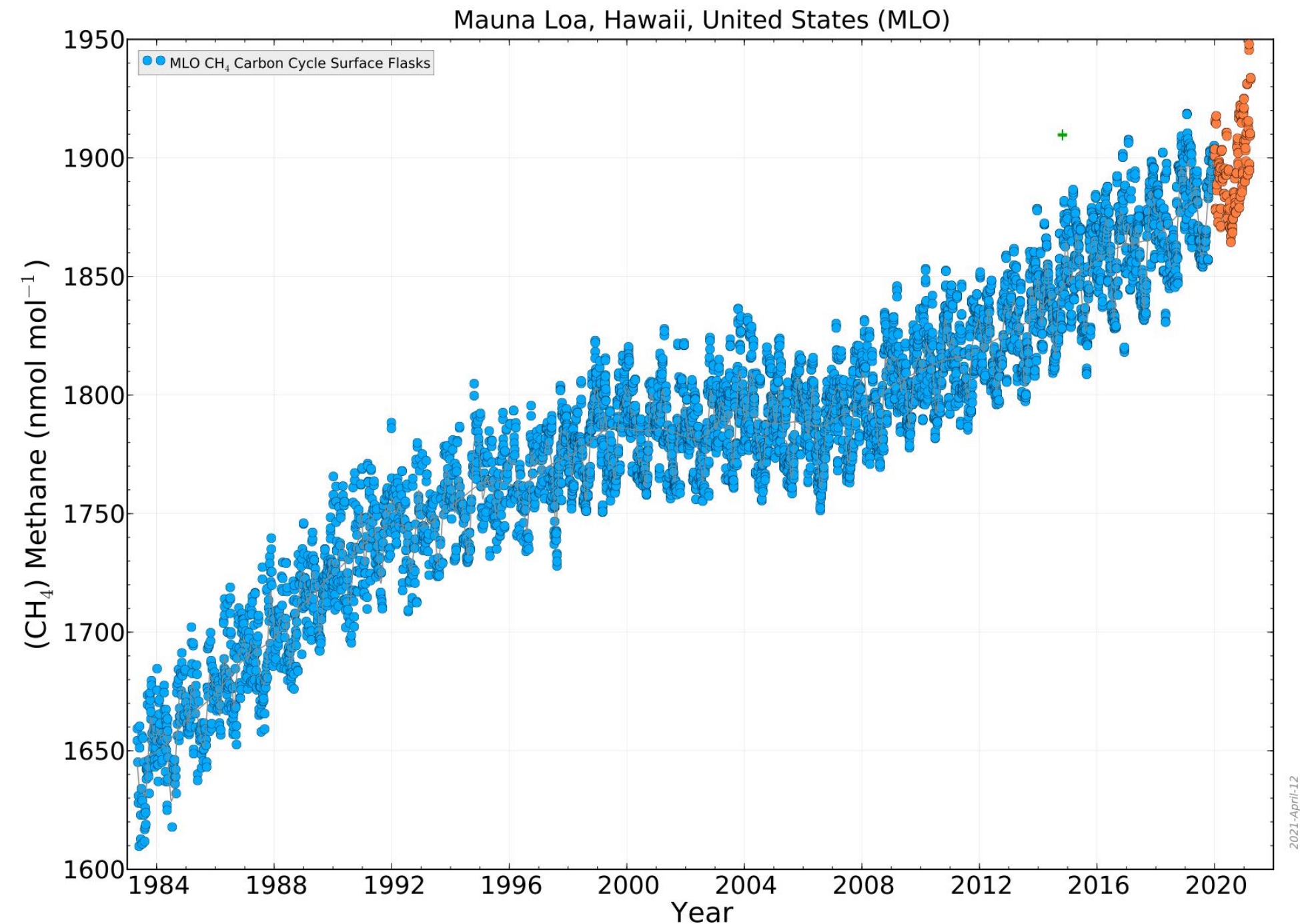
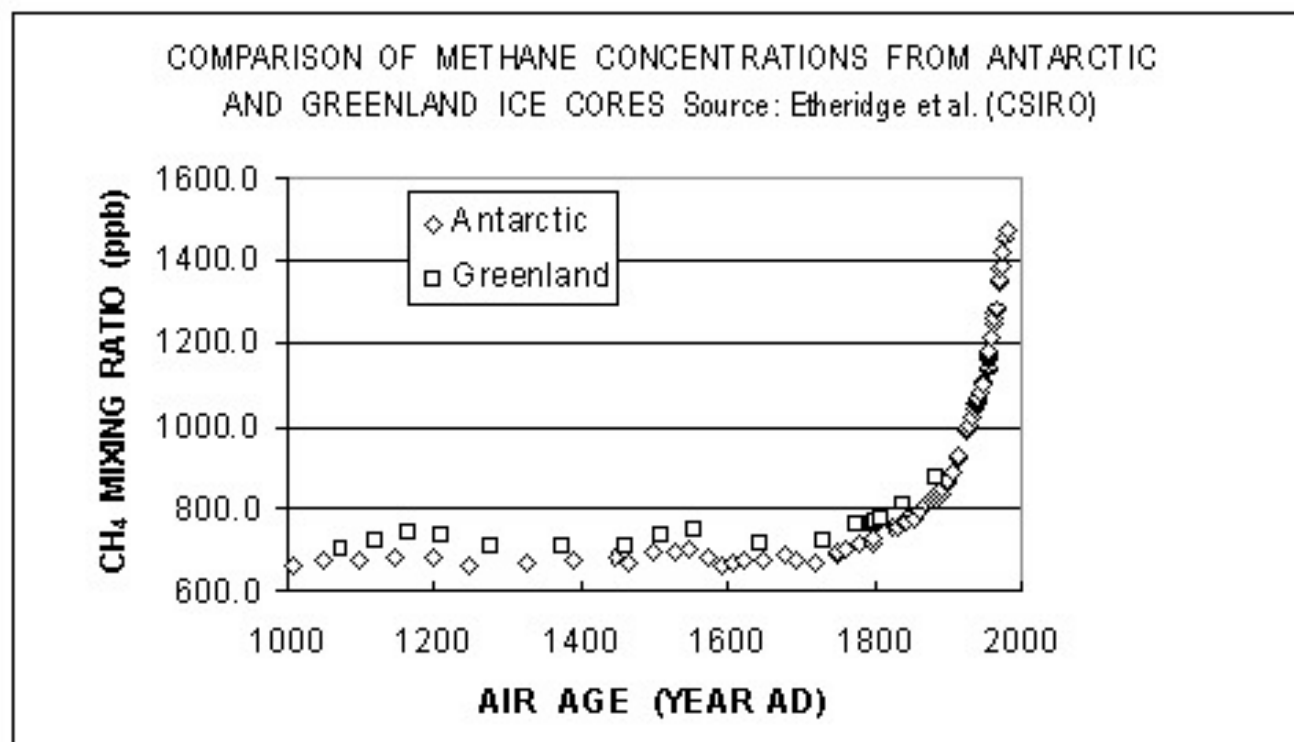
Methane (CH₄)

Produced in

- wetlands
- various human activities e.g. energy industry, rice paddies, cows

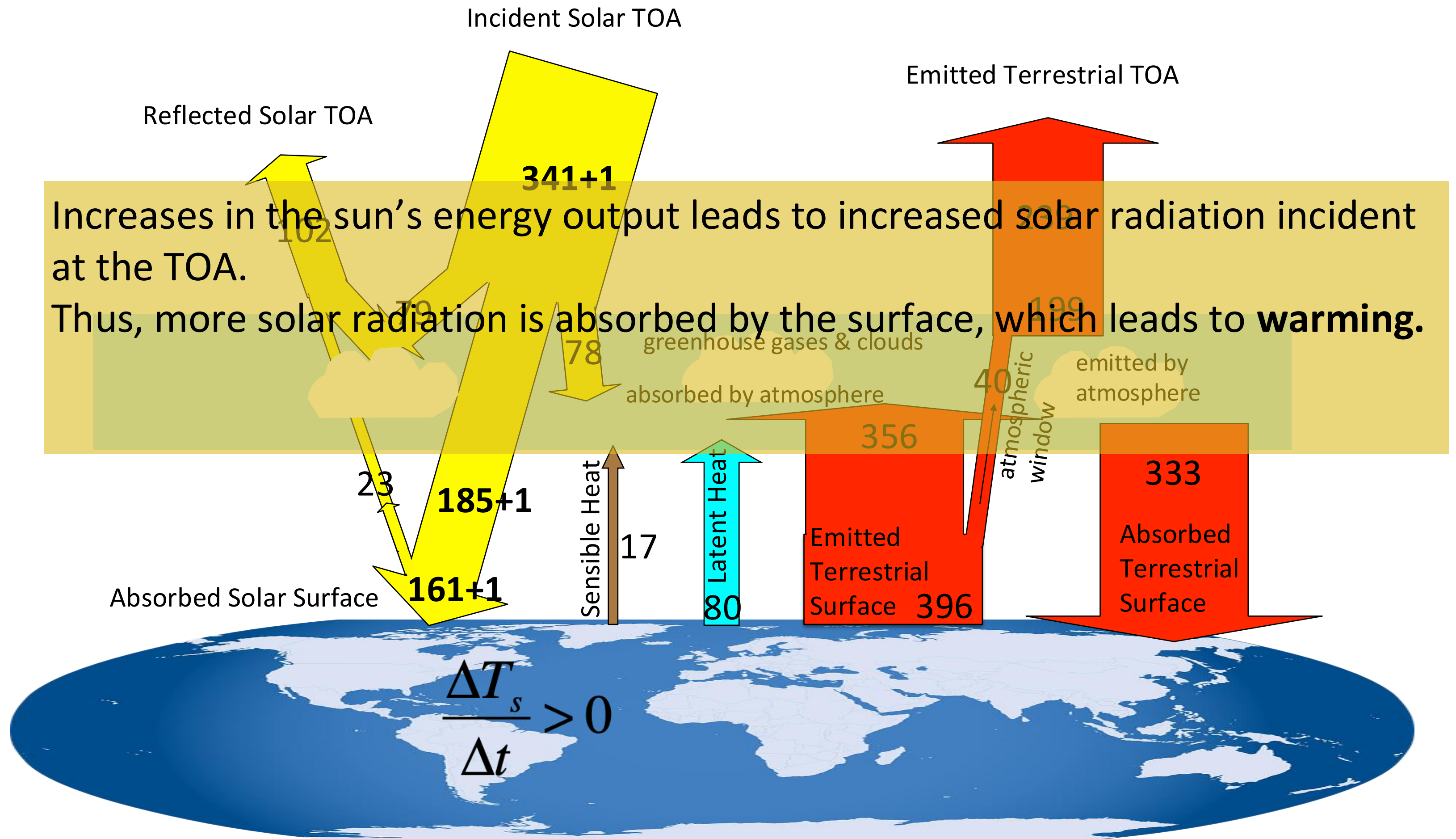
Removed by chemical reaction with OH

Lifetime ~ 12 years

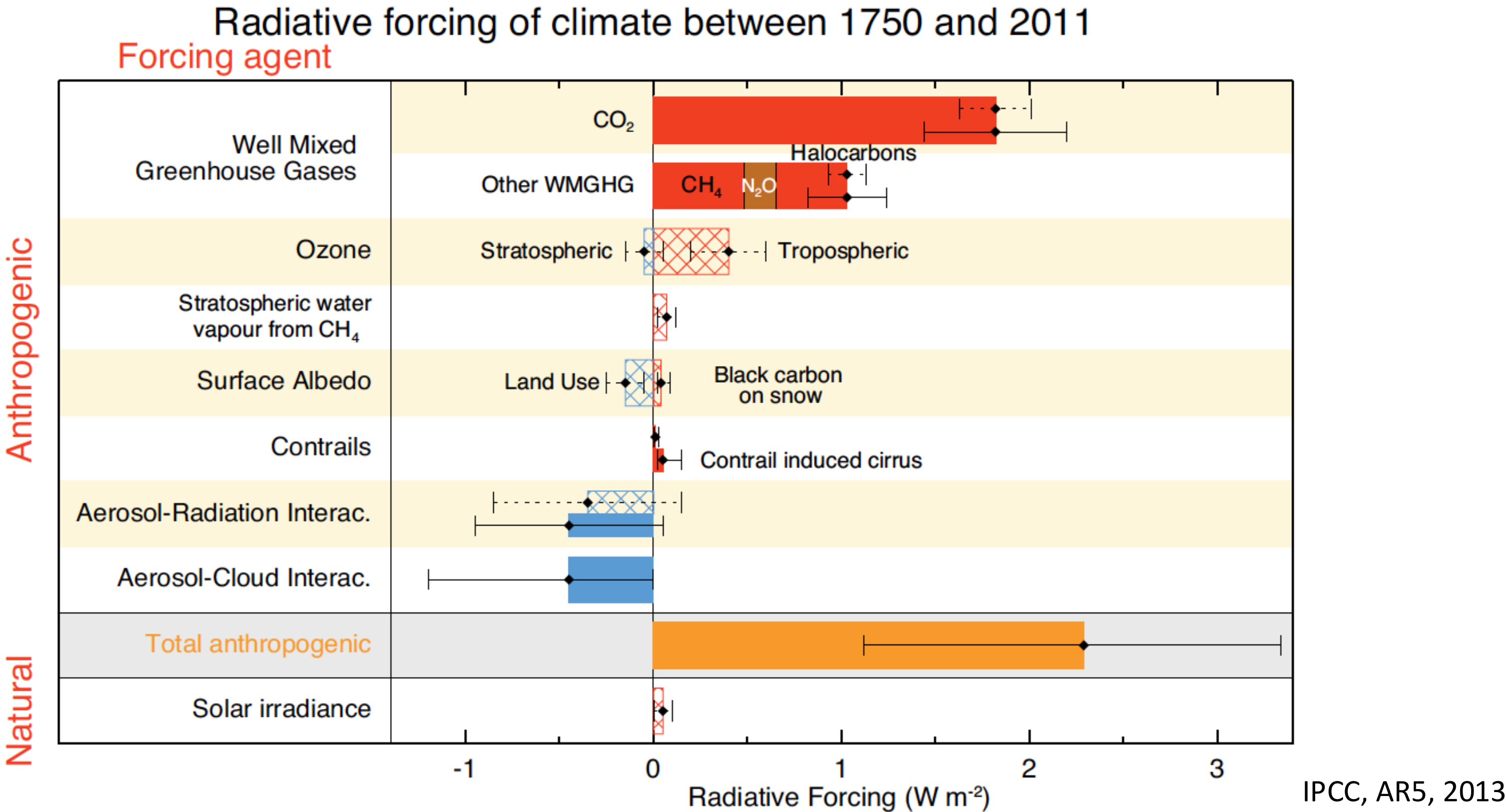


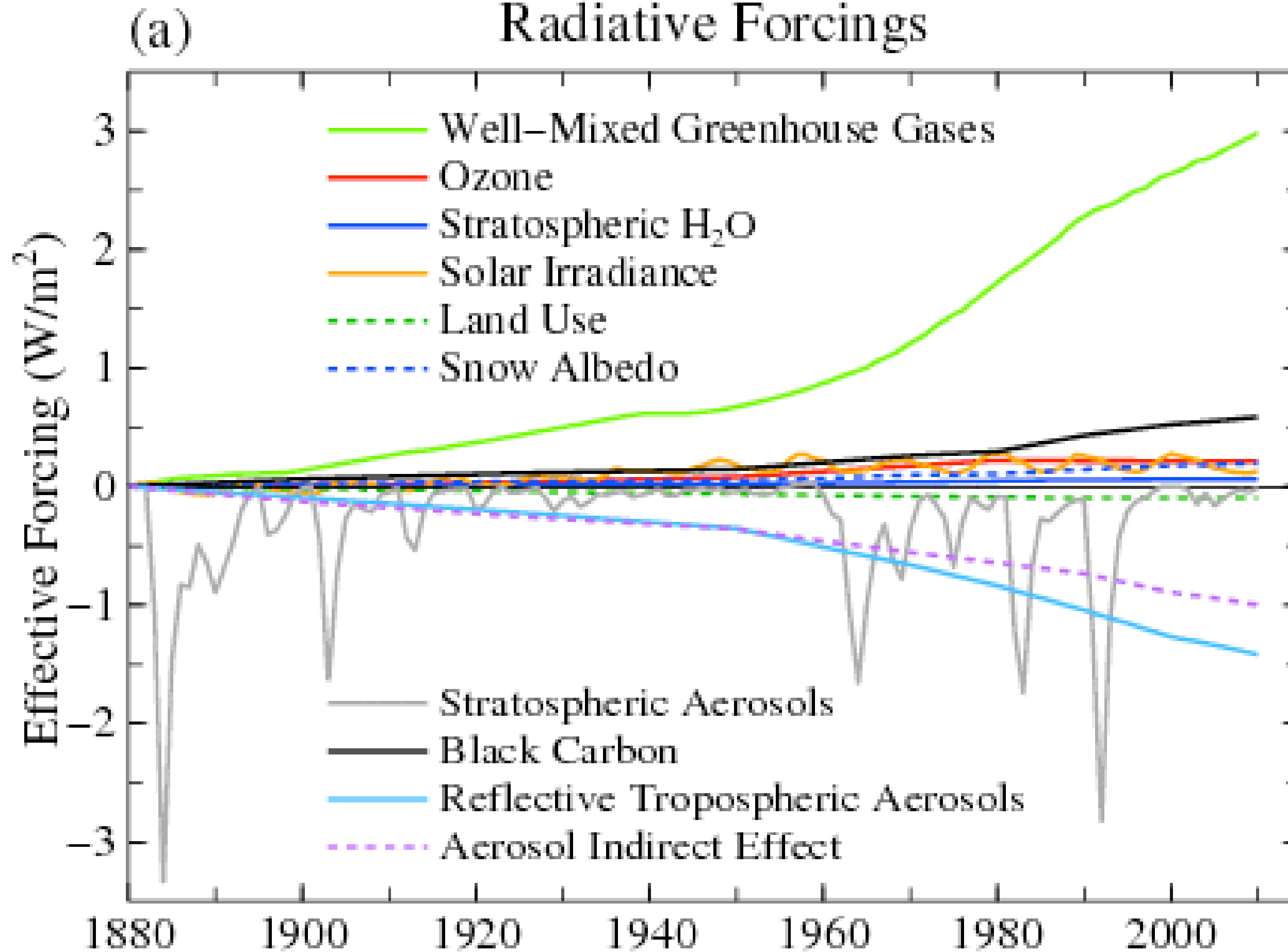
Radiative Forcing 1750-2011: <http://www.esrl.noaa.gov/gmd/dv/iadv/>
 $\Delta F_{\text{CH}_4} = 0.5 \text{ W/m}^2$

http://cdiac.ornl.gov/trends/atm_meth/lawdome_meth-graphics.html



Global, mean radiative forcings of important atmospheric agents over the period 1750-2005



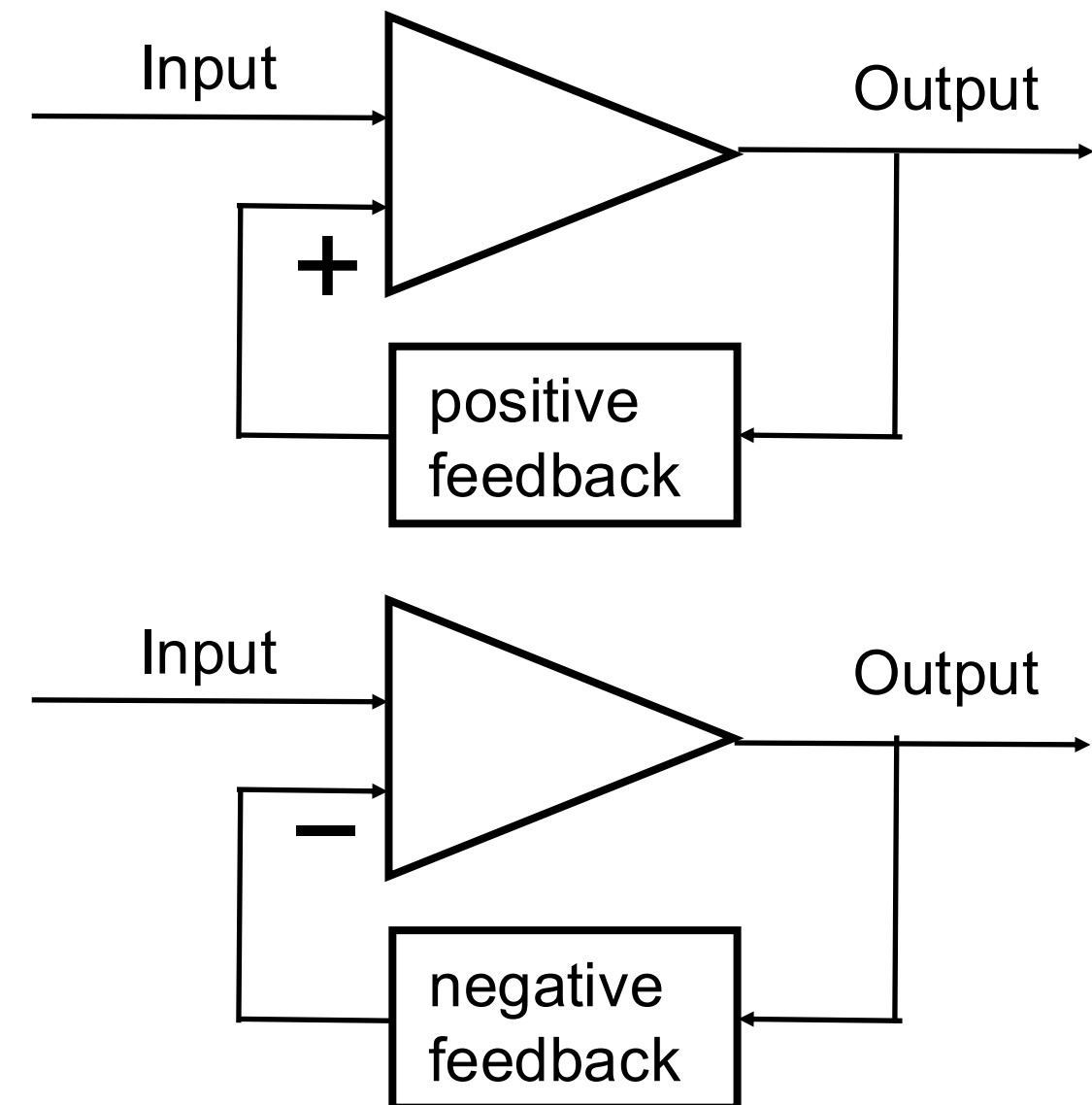


Climate Feedbacks: Modifiers of Climate Change

A feedback is a process that either amplifies or dampens the response of the climate system to a forcing.

Sign convention:

- A positive feedback amplifies the signal.
- A negative feedback dampens the signal.



Positive Feedback Example

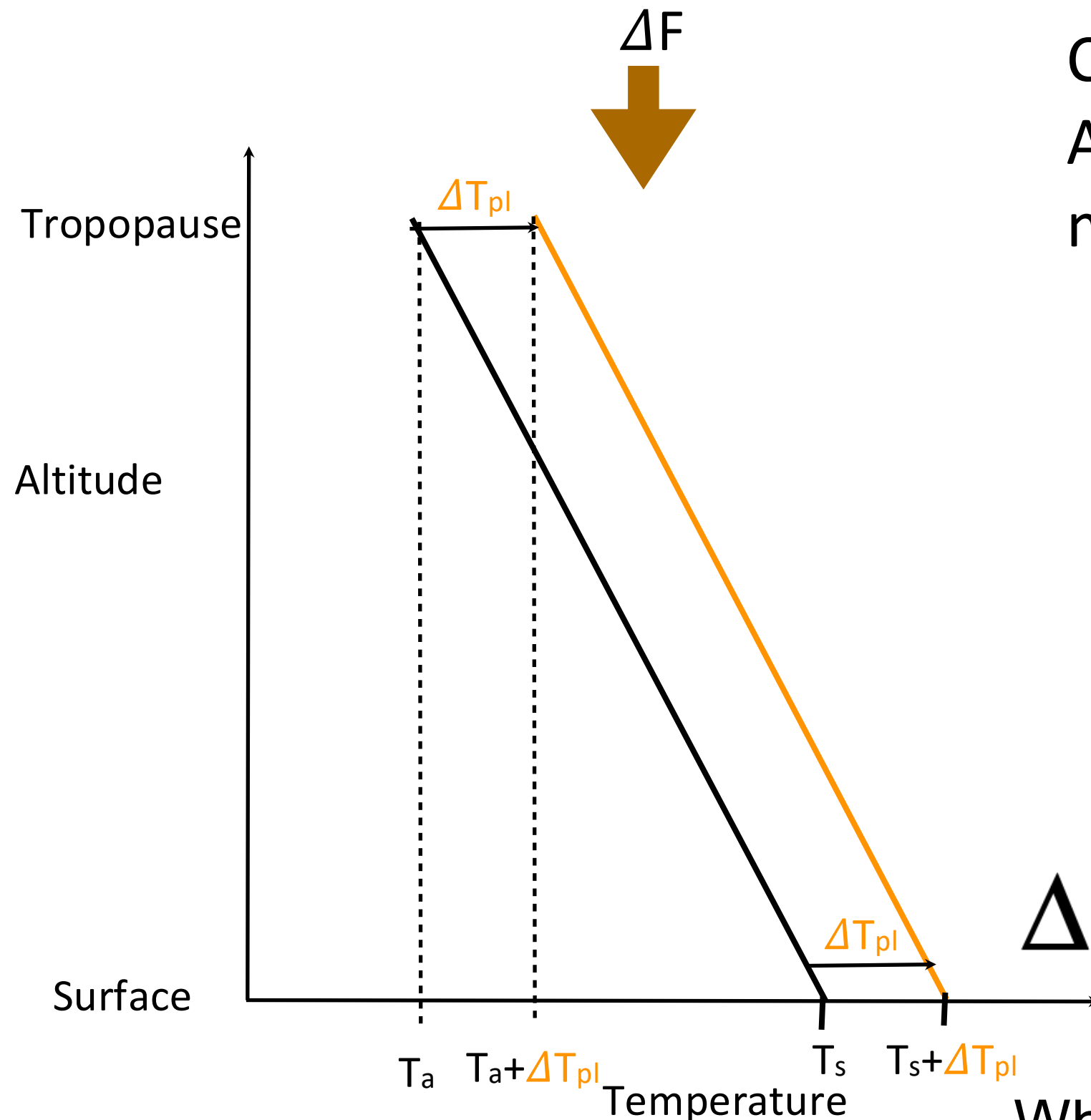
- Your talking into a microphone
- An amplifier makes your voice louder
- A positive feedback can lead to a runaway effect if no or only weak negative feedbacks are involved (you hold the microphone in front of the speaker).
Snowball Earth.

Negative Feedback Example

- You're talking into a pillow
- This makes your voice quieter
- A negative feedback is stabilizing, it prevents a runaway effect

Initial change in temperature due to radiative forcing

Assume positive radiative forcing ΔF (e.g. increased CO_2). This leads to a surface warming.
Assume warming is uniform (troposphere is well mixed).



input = output
 $(1 - a)S = \sigma T_a^4$

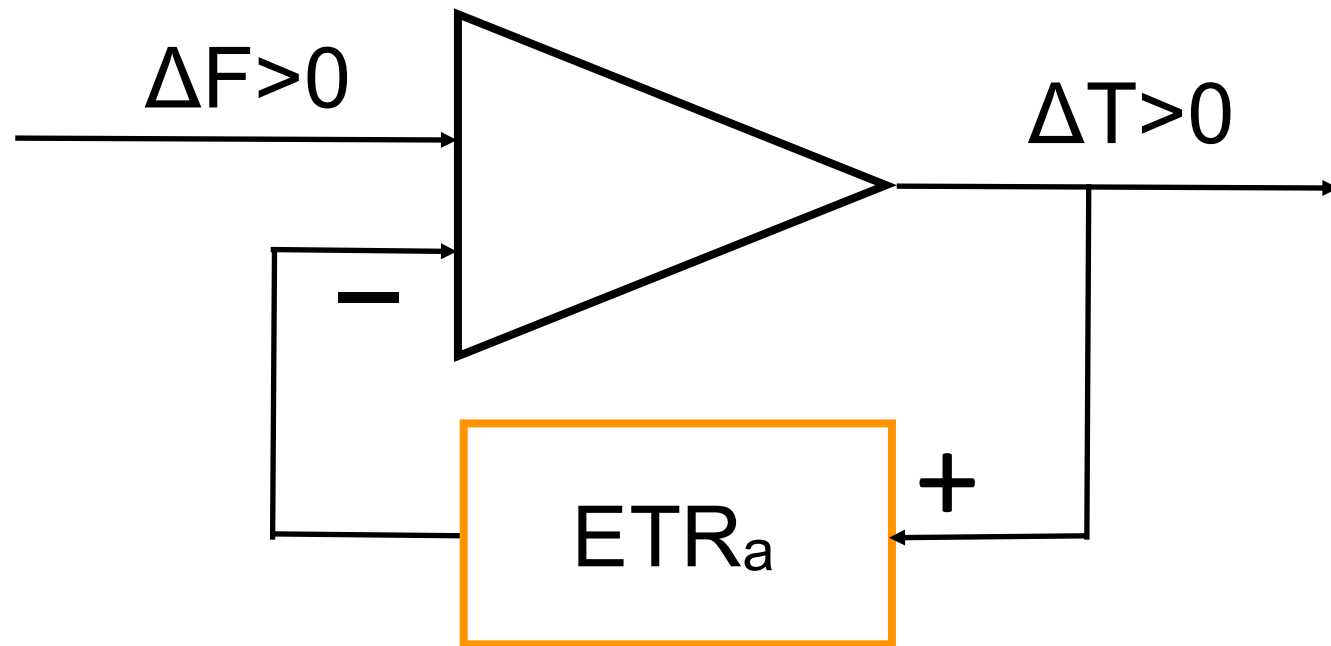
$$(1 - a)S + \Delta F = \sigma (T_a + \Delta T_{pl})^4$$

This leads to increased outgoing
longwave radiation.
(Planck feedback)

$$\Delta T_{pl} = ([(1 - a)S + \Delta F] / \sigma)^{1/4} - T_a$$

What is the temperature change for a doubling of CO_2 ?

Planck Feedback



- Positive forcing leads to warming,
- which leads to increased emitted terrestrial radiation

$$ETR_a = \sigma (T_a + \Delta T_{pl})^4$$

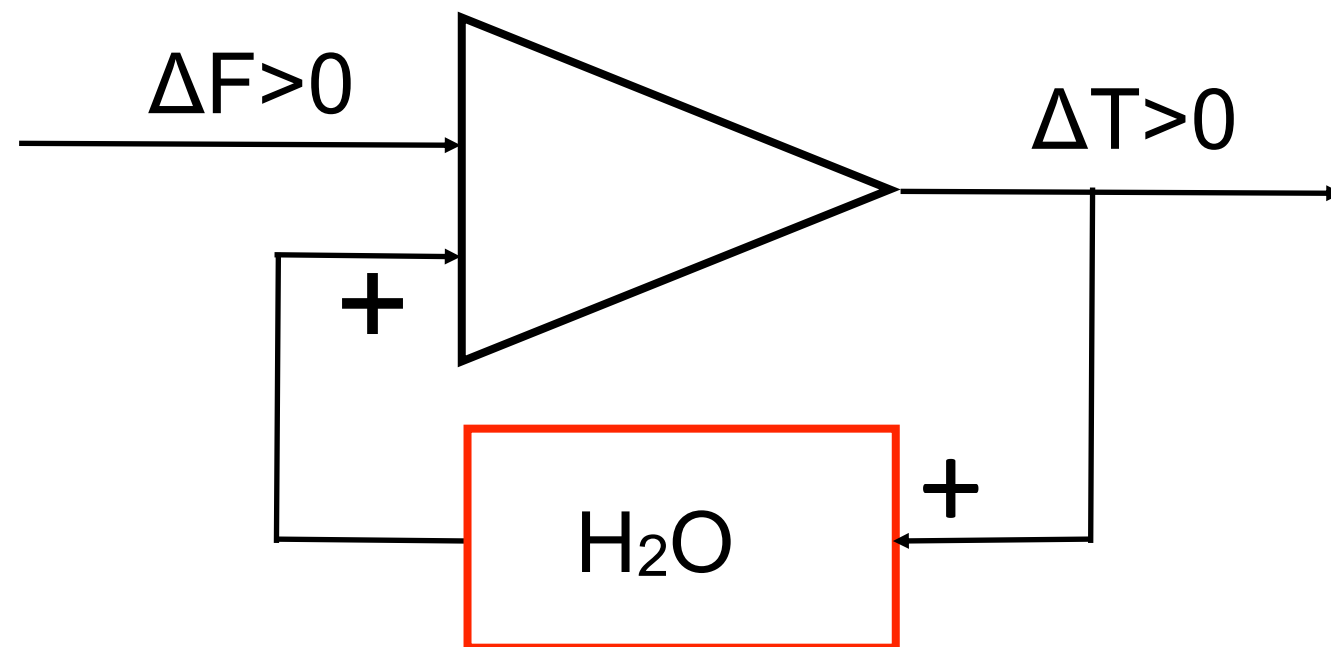
- this cools the Earth
- => negative feedback

$$\Delta T_{pl} = ([(1 - a) S + \Delta F] / \sigma)^{1/4} - T_a$$

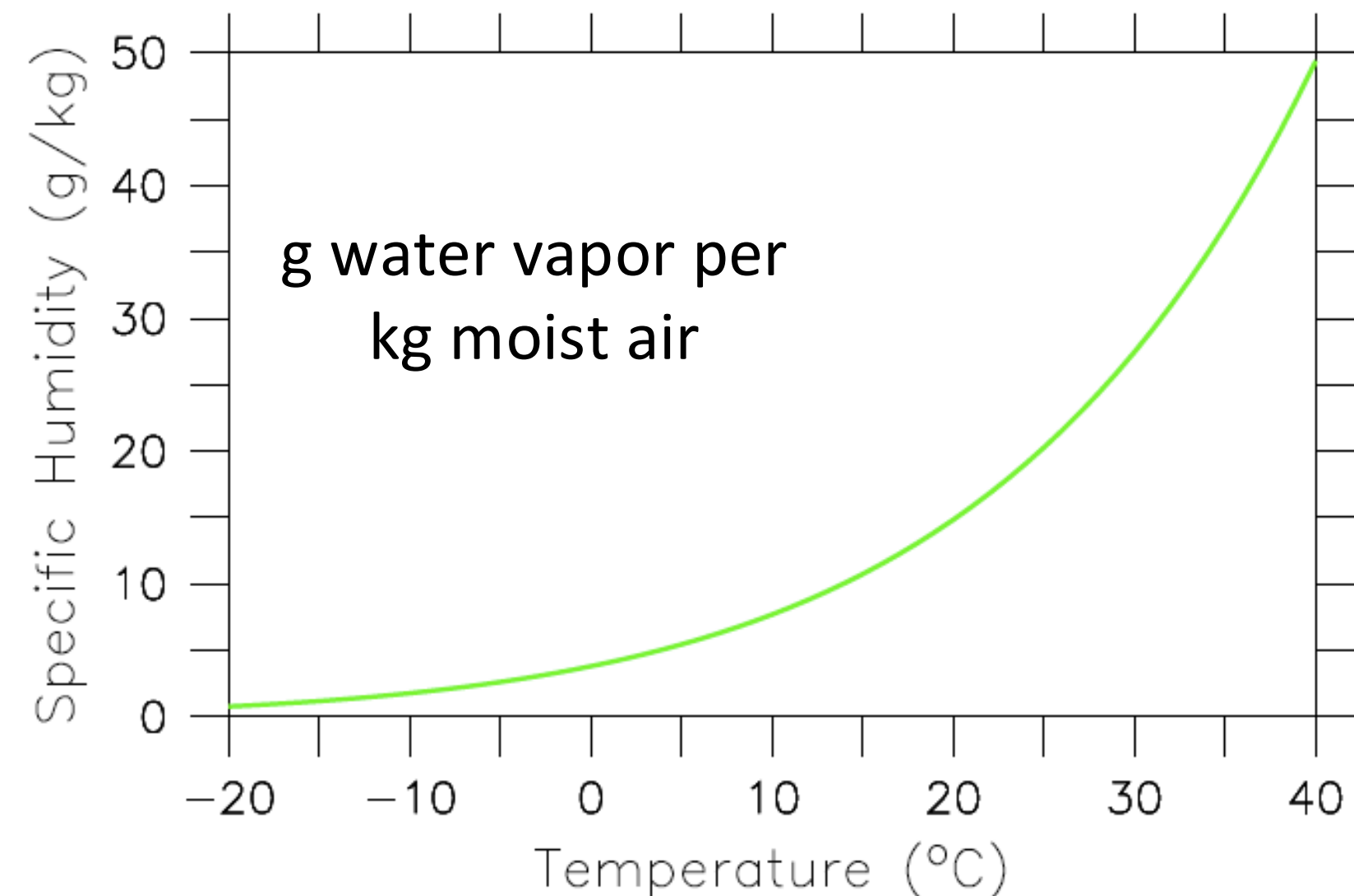
For doubling of CO_2 $\Delta F = 3.7 \text{ Wm}^{-2}$: $\Delta T_{pl} = 1 \text{ K}$

Water Vapor Feedback

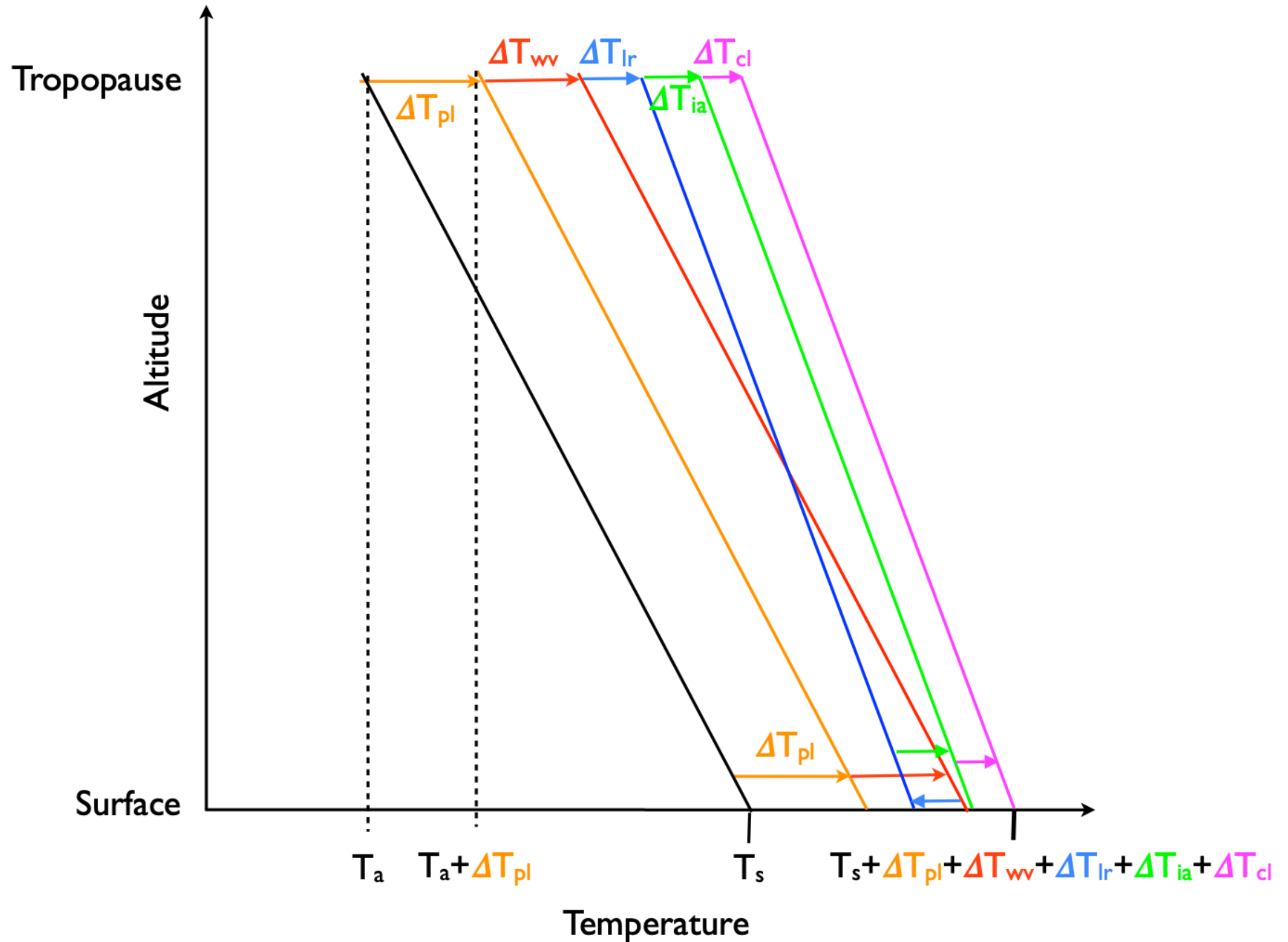
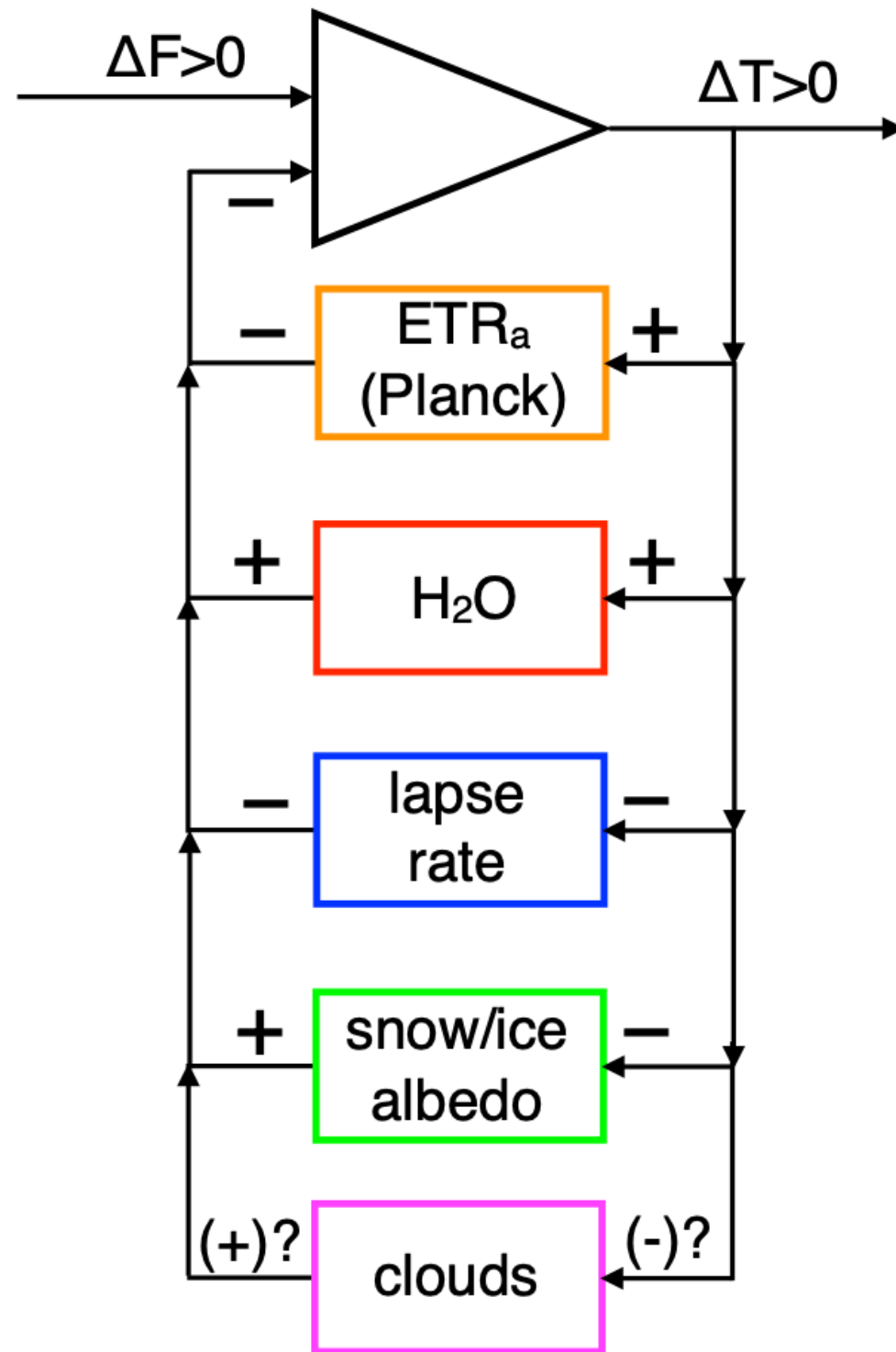
- assume a positive forcing leads to surface warming
- surface warming leads to more evaporation and water vapor in the air (Clausius-Clapeyron equation)
- increased water vapor in atmosphere leads to additional warming
- => positive feedback



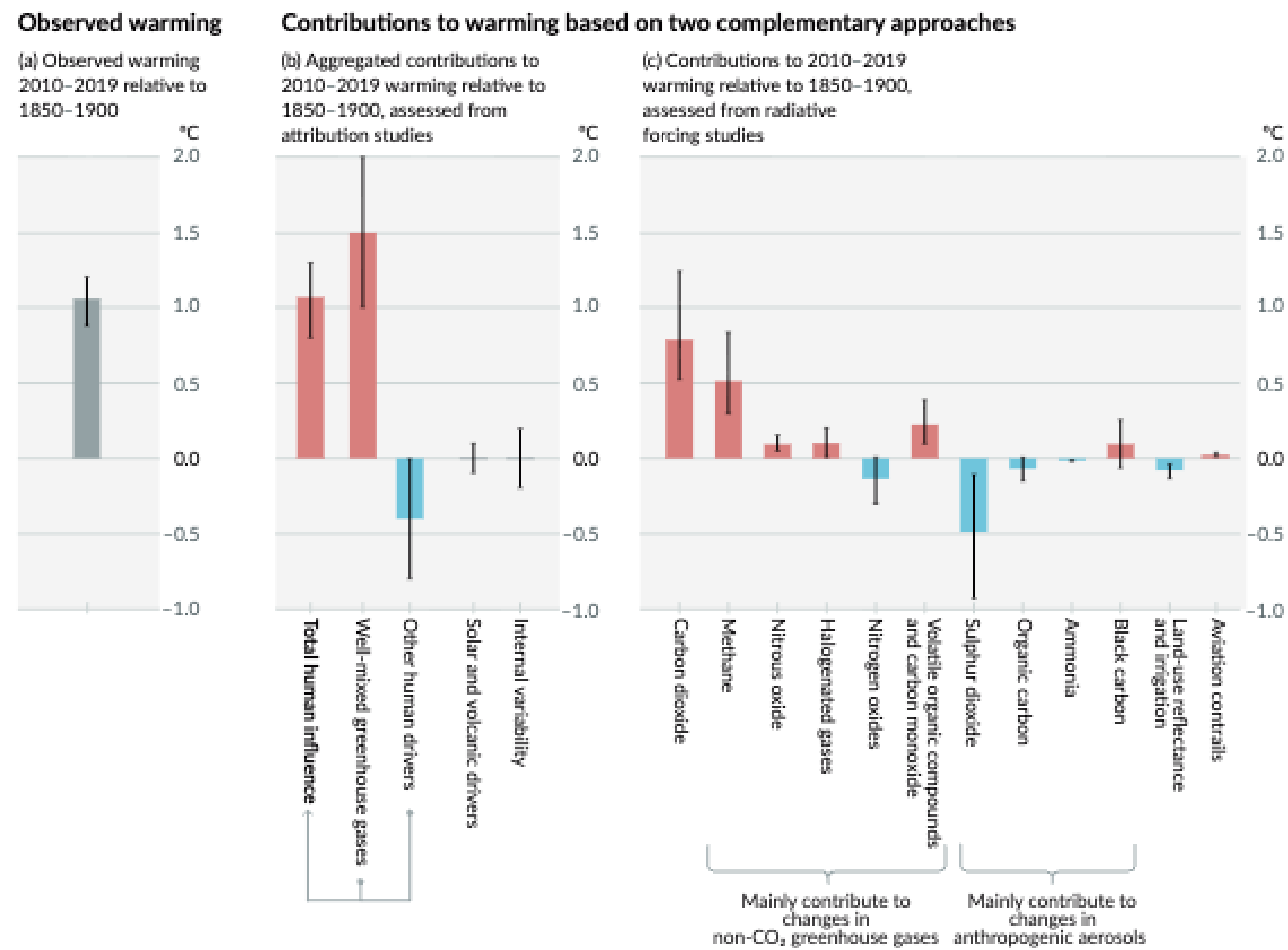
30 g of water = 30 mL = 1 us fl oz



All Feedbacks



Observed warming is driven by emissions from human activities, with greenhouse gas warming partly masked by aerosol cooling



- Climate feedbacks can amplify (+) or dampen (-) the initial climate change caused by the forcing
- Planck (-), water vapor (+), lapse rate (-), ice albedo (+), clouds (+)
- Climate sensitivity is the temperature change for a given forcing (e.g. doubling of CO₂)

Climate Sensitivity

Climate sensitivity ΔT_{2x} is the global mean near surface temperature change **at equilibrium** for a doubling of atmospheric CO₂ (including feedbacks)

Sometimes the climate sensitivity is also reported in units of K/(W/m²). In this case it is the temperature change divided by the forcing $S_C = \Delta T / \Delta F$. Because we know the forcing for a doubling of CO₂, $\Delta F_{2x} = 3.7 \text{ W/m}^2$, S_C can be calculated from ΔT_{2x} or vice versa: $S_C = \Delta T_{2x} / \Delta F_{2x}$.

Climate sensitivity is the result of all (fast) feedbacks. Its value is most likely between about 1.5 K and 4.5 K, with a best estimate of about $\Delta T_{2x} = 3 \text{ K}$. Its uncertainty is due to uncertainties in the feedbacks, with the cloud feedback being the least well known.

Feedback Parameters

Feedback parameters $\gamma_i = (\Delta R / \Delta x_i) \times (\Delta x_i / \Delta T)$ describe how the radiative fluxes at the top-of-the-atmosphere R respond to one controlling variable x_i times the sensitivity of that variable to a global temperature change. They are useful to quantitatively compare different feedbacks.

Planck: $\gamma_{pl} = -3.2 \text{ (W/m}^2\text{)/K}$

Water Vapor + Lapse Rate: $\gamma_{wv} + \gamma_{lr} = 1.5 - 0.5 = +1 \text{ (W/m}^2\text{)/K}$

Ice Albedo: $\gamma_{ia} = +0.3 \text{ (W/m}^2\text{)/K}$

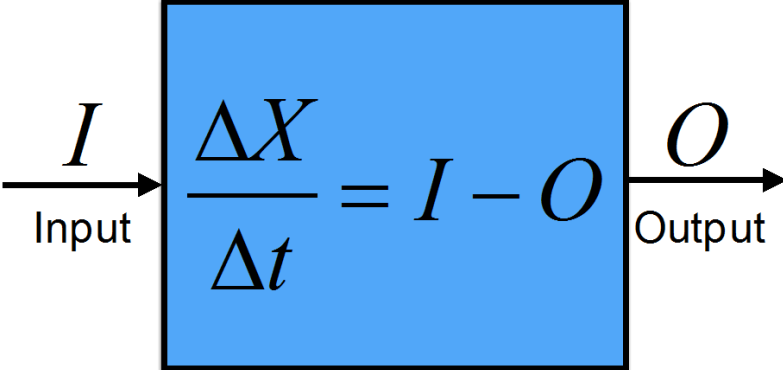
Clouds: $\gamma_{cl} = +0.8 \text{ (W/m}^2\text{)/K}$

Total Feedback: $\gamma = \gamma_{pl} + \gamma_{wv} + \gamma_{lr} + \gamma_{ia} + \gamma_{cl} = -1.1 \text{ (W/m}^2\text{)/K}$

Climate Sensitivity: $S_c = -1/\gamma$

Non Equilibrium Response

Earth's Energy Budget Equat



The diagram shows a blue square box representing a system. An arrow labeled I with the word "Input" below it points into the left side of the box. Inside the box, the equation $\frac{\Delta X}{\Delta t} = I - O$ is written. An arrow labeled O with the word "Output" below it points out of the right side of the box.

$$C \frac{\Delta T}{\Delta t} = \underbrace{(1 - a)S - \sigma T^4}_R$$

Rate-of-Change of Heat Content = Energy Input - Energy Output

Heat Content = Heat Capacity (C) times Temperature (T)

At Equilibrium

$$C \frac{\Delta T}{\Delta t} = (1 - a)S - \sigma T_{eq}^4 = 0$$

Add Forcing (instantaneously)

$$C \frac{\Delta T}{\Delta t} = (1 - a)S - \sigma T_{eq}^4 + \Delta F$$

How will that affect temperature?

$$C \frac{\Delta T}{\Delta t} > 0$$

It leads to warming.

How fast would that warming be?

Add Forcing (instantaneously)

$$C \frac{\Delta T}{\Delta t} = \Delta F$$

If radiative fluxes at TOA wouldn't change

$$\frac{\Delta T}{\Delta t} = \frac{\Delta F}{C}$$

Heat Capacity:

Assume $H_O = 50$ m ocean mixed layer depth

$H_A = 8300$ m height of atmosphere

Mass per area of atmosphere

Mass per area of ocean

Specific heat

$$m_A = H_A \rho_A = 1 \times 10^4 \text{ kg/m}^2$$

$$m_O = H_O \rho_O = 5 \times 10^4 \text{ kg/m}^2$$

$$C_{Ap} = 1,000 \text{ J/(kgK)} \quad C_{Op} = 4,200 \text{ J/(kgK)}$$

$$C = C_A + C_O = m_A C_{Ap} + m_O C_{Op} = 2.2 \times 10^8 \text{ J/(m}^2 \text{K)}$$

$$C_O = 20 \times C_A$$

Calculate the rate of temperature change for instantaneous doubling of CO_2 !

Earth's Energy Budget Equation

Add Forcing

$$\frac{\Delta T}{\Delta t} = \frac{\Delta F}{C} = 1.7 \times 10^{-8} K/s = 0.54 K/yr$$

How does this compare to our current rate of change?

For a time step $\Delta t = 1$ year we get:

Year 1: $\Delta T_{y1} = (\Delta F/C) \times \Delta t = 0.54 \text{ K}$

Earth's Energy Budget Equation

Add Forcing

$$C \frac{\Delta T}{\Delta t} = (1 - a)S - \sigma T_{eq}^4 + \Delta F$$

How will that affect temperature?

$$C \frac{\Delta T}{\Delta t} > 0$$

It leads to warming.

What else will happen?

Earth's Energy Budget Equation

Add Forcing

$$C \frac{\Delta T}{\Delta t} = \underbrace{(1 - a)S - \sigma T^4}_R + \Delta F$$

Warming will lead to changes in TOA radiative fluxes R

We can approximate a response of the outgoing longwave radiation to the 0.5 K warming in year 1 calculated earlier. (Let's assume the shortwave doesn't change for now. We also assume that in year 1 R hasn't responded to any warming.)

Earth's Energy Budget Equation

Add Forcing

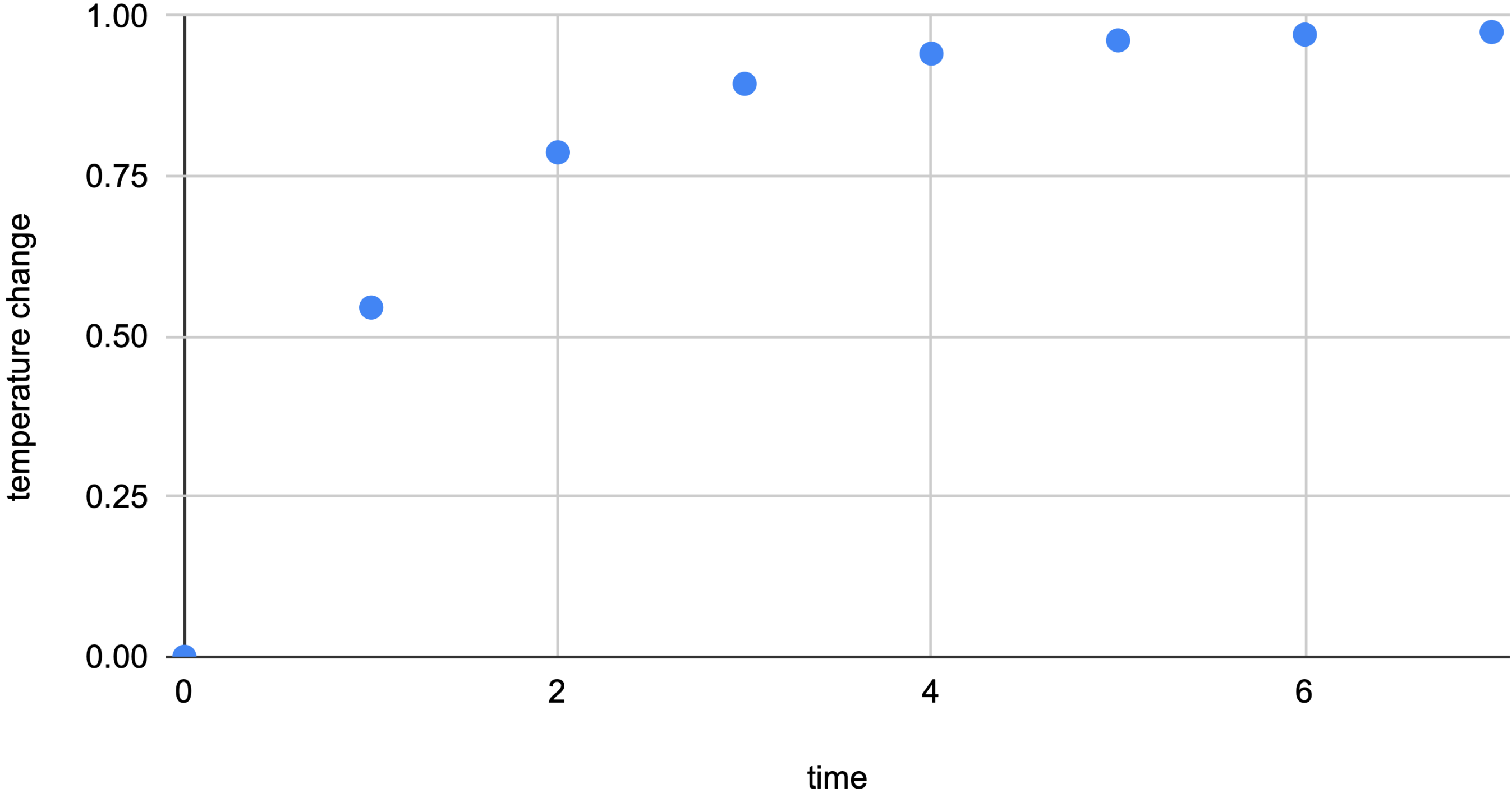
$$C \frac{\Delta T}{\Delta t} = (1 - a)S - \sigma(T_{eq} + \Delta T_{y1})^4 + \Delta F$$

In year 2 $R = (1 - a)S - \sigma(T_{eq} + \Delta T)^4 = -1.9 \text{ W/m}^2$

$$\frac{\Delta T}{\Delta t} = \frac{R + \Delta F}{C} = (3.7 - 1.9) \text{ W/m}^2 / C = 0.26 \text{ K/yr}$$

You can code this in a Google spreadsheet

temperature change vs. time



What happens if we increase the heat capacity?

temperature change vs. time

